Routing Protocol Configuration -- Guided Notes

Unit 2.4.1 | CompTIA Network+ N10-009 Obj. 2.4 | N10-008 2.4

Name: ___________________________________ Date: ____________ Period: ______

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

A _______ route is manually configured by an administrator and does not update automatically when the network changes. A _______ route is learned automatically through a routing protocol. The _______ route is used when no more specific match exists in the routing table.

Question 2

Distance-vector routing protocols share their _______ table with directly connected neighbors at regular intervals. The two classic distance-vector protocols are _______ (used in small networks) and _______ (used in large enterprise networks).

Question 3

OSPF is a _______ -state routing protocol. Instead of sharing routing tables, routers share _______ advertisements (LSAs) to build a complete map of the network called the _______. Each router then independently calculates the best path using Dijkstra's algorithm.

Question 4

Administrative distance (AD) is used when a router learns about the same network from _______ sources. The route with the _______ AD value is preferred. List the AD values for: directly connected (___), static route (___), OSPF (___), RIP (___).

Question 5

BGP (Border Gateway Protocol) is the routing protocol that runs the _______. It is classified as an _______ gateway protocol because it routes between different autonomous systems. Unlike OSPF or EIGRP, BGP makes routing decisions based primarily on _______.

Question 6

Route redistribution allows a router to take routes learned from one protocol (e.g., OSPF) and advertise them into another (e.g., EIGRP). Why might this be necessary, and what is one risk of misconfigured redistribution?

Question 7

In OSPF, routers within the same ________ exchange LSAs directly. Area 0 is called the ________ area and all other areas must connect to it. This hierarchical design reduces the size of the ________ on each router.

Question 8 -- Real World Application

REAL WORLD: A network engineer notices that traffic between two sites is taking an unexpected path -- traveling through a slower WAN link instead of a faster fiber connection. Both paths are learned via OSPF. What OSPF value would the engineer adjust to prefer the faster link, and how does OSPF calculate this value by default?

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/2-4-1-routing-protocol-configuration/2-4-1-routing-protocol-configuration-answer-key.pdf
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